Sharon Kodali

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Education

University of California – San Diego — San Diego, CA

B.S. Data Science, Minor in Cognitive Science (Machine Learning), GPA: 4.00 — *Expected 2028*

Relevant Courses: Data Structures & Algorithms, Systems Programming, Linear Algebra, Probability & Statistics

Experience

Undergraduate Research Intern — Nov 2026 – Present

San Diego Supercomputer Center (SDSC) — *La Jolla, CA*

- Porting a C++ fusion plasma simulation to JAX with Dr. Igor Sfiligoi for portable GPU execution (XLA/`jit`) and automatic differentiation (`grad`) for reactor-design optimization
- Building the port's regression suite: reading C++ routines to extract inputs, outputs, invariants and edge cases, then writing parametrized pytest tests against C++ reference outputs with `rtol`/`atol` tolerances
- Validating AI-generated and hand-written JAX kernels for numerical equivalence with the C++ originals

Software Development Intern (iOS) — Jul 2026 – Present

Rydr — *Remote*

- Cut API response time 62% (800ms to 300ms) by implementing local caching for a ride-sharing app
- Caught 12 bugs pre-release by writing 30+ integration tests for the payment flow (85% coverage)

Data & Machine Learning Analyst — May 2026 – Aug 2026

Sprout Media — *Remote*

- Cut pipeline runtime 40% (60 to 36 min) by refactoring batch API calls into async requests; cut null and duplicate records 25% with validation checks

Project Manager & Developer — Nov 2024 – Jun 2025

Open Coding Society — *San Diego, CA*

- Owned a Java (Spring Boot) and MySQL platform with REST APIs on AWS, used by 500+ students at 5 high schools, from data model to UI
- Cut bug turnaround 60% (5 to 2 days) by leading sprint planning, demos and code reviews (40+ PRs) for 30 engineers

Projects

Distributed Key-Value Store | *Go, Raft, Docker* — Code

- Built a 5-node fault-tolerant Raft cluster in Go, achieving zero data loss and leader re-election in under 4s after killing the leader mid-write
- Sustained 3,200 writes/sec at 500 concurrent clients and sub-3ms p99 reads; wrote 100+ tests (89% coverage) under Go's race detector
- Fixed 2 distributed-systems bugs (an IP/hostname mismatch and a DNS startup race) that surfaced only under real network conditions; containerized with Docker

TritonETA – Real-Time Shuttle Tracker | *Python, SQL, DuckDB, dbt* — Code

- Built an automated ingestion pipeline polling a live GTFS-RT GPS feed every 15–30s via GitHub Actions, collecting 180K GPS pings from 28 shuttles unattended
- Modeled the data in 16 version-controlled dbt models with 28 data tests (including a train/test leakage test) and documented lineage

ResumeTailor – AI Resume Tailoring Platform | *FastAPI, Next.js, OpenAI API* — Code | Demo

- Built a full-stack LLM app that tailors resumes to job postings by rewriting LaTeX bullets in place via offset splicing, so document structure never breaks
- Added a relevance-scoring metric to check model output before trusting it; covered the backend with 96 pytest tests and CI

Technical Skills

Languages: Go, C/C++, Python, Java, SQL, JavaScript, TypeScript, Bash
Systems: Distributed Systems, Raft Consensus, Concurrency, Networking, Docker, REST APIs, AWS
Data & Storage: PostgreSQL, MySQL, DuckDB, dbt, Data Structures & Indexing
Tools: Git, CI/CD (GitHub Actions), pytest, JAX, FastAPI, Next.js, Spring Boot, Claude Code, Cursor